

Problem Statement

Frontline health workers in low-resource African communities frequently need to interpret clinical symptoms and make treatment decisions without immediate access to a physician, specialist, or up-to-date medical literature.

This project addresses that gap by developing a domain-specific small language model (SLM) for medical advice and clinical decision support. The approach centers on building a clinically validated dataset curated from peer-reviewed journals, clinical guidelines, and anonymized healthcare records, with data accuracy, patient privacy, and medical ethics treated as first-order design constraints rather than afterthoughts.

The model will be fine-tuned and equipped with adequate guardrails to prevent hallucination and keep the outputs evidence-based.

This SLM would demonstrate that a compact, efficiently trained model could give frontline workers a reliable second opinion in moments that matter, helping close the gap in care access without compromising patient wellbeing or trust.